Your First AI Workflow

Connecting the OpenAI API
Week 2: Designing AI Assistants & Agents

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Session Topics

- ✔ What a credential is
- ✔ Getting an OpenAI API key
- ✔ Adding the credential in n8n
- ✔ Building the workflow
- ✔ Reading the output
- ✔ Costs and common errors

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What a Credential Is

- A credential is how n8n proves to another service that it is allowed to act on your behalf.
- You add it once and every node that talks to that service can reuse it.
- The key is stored separately, so it never sits inside the workflow you might share or export.
- Treat an API key like a password. Never paste it into chat, a slide or a shared document.

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Getting an OpenAI API Key

1. Sign in at platform.openai.com. This is separate from a ChatGPT subscription and billed separately.
2. Add a small amount of credit under Billing. API use is pay as you go, not a monthly plan.
3. Open the API keys page and create a new secret key.
4. Give it a name you will recognise later, such as n8n-training.
5. Copy the key straight away. It is shown once and cannot be viewed again afterwards.
6. Store it in a password manager, not in a text file on your desktop.

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Adding the Credential in n8n

1. In n8n, open Credentials and choose to create a new one.
2. Search for OpenAI and select it from the list.
3. Paste the key into the API key field and save.
4. Wait for the connection check. Fix a red error here rather than pushing on.
5. The credential now appears in a dropdown inside every OpenAI node you add.

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Building the Workflow

1. Start a new workflow and add a manual trigger so you can run it whenever you like.
2. Add an OpenAI node and pick the credential you just created.
3. Choose the operation, select a model, and write your prompt in the message field.
4. Keep the first prompt to one clear instruction. Complexity comes later.
5. Add one node after it that does something with the answer, or leave it and just read the output.

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Reading the Output

- Press execute. A green tick on every node means the run reached the end.
- Click any node to see its input on one side and its output on the other.
- The model answer sits inside a field in the output data. Expand it and find that field.
- You refer to that field by name in later nodes. This is how the answer gets passed along.

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Writing the Prompt Inside the Node

- Be specific about the job: summarise in three lines, sort into one of these categories, pull out these fields.
- Say what to do when the model is unsure, so it does not fill the gap with an invented answer.
- If a later node has to read the result, ask for a fixed shape rather than free text.
- Test the same prompt against two or three different inputs before you trust it.

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Costs and Common Errors

- You pay per run, based on how much text goes in and comes out. Classroom testing costs very little.
- **A 401 error:** the key is wrong or was not saved. Create the credential again.
- **A 429 error:** you are out of credit or sending requests too fast. Check billing first.
- **Empty output:** usually a blank prompt field, or a reference to a field that does not exist.

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What You Can Do Now

- Create an OpenAI credential in n8n, and explain why the key lives outside the node.
- Build a workflow of three nodes: a trigger, a model, and something that uses the answer.
- Run it, open the output and find the model response in the data.
- Next: prompt engineering inside n8n, starting with system and user messages.

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